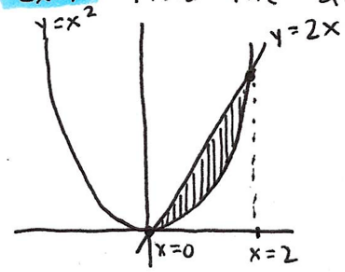
Session 1 : 1/29/24

Textbook : 6.1 Area between curves

ex 1: Find the area of the region bounded by $y = 2x$ and $y = x^2$.



Intersect when

$x^2 = 2x$

$x^2 - 2x = 0$

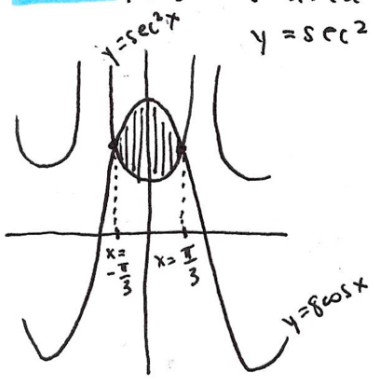
$x(x-2) = 0$

$x = 0, 2$

$$A = \int_a^b [f(x) - g(x)]\,dx$$
$$= \int_0^2 (2x - x^2)\,dx$$
$$= \left(2 \cdot \frac{x^2}{2} - \frac{x^3}{3}\right)\Big|_0^2$$
$$= 4 - \frac{8}{3}$$
$$= \boxed{\frac{4}{3}}$$

ex 2: Find the area of the region bounded by the given curves: (6.1 #29)

$y = \sec^2 x, \quad y = 8\cos x, \quad -\frac{\pi}{3} \le x \le \frac{\pi}{3}$



Intersect when

$\sec^2 x = 8\cos x$

$\frac{1}{\cos^2 x} = 8\cos x$

$\frac{1}{8} = \cos^3 x$

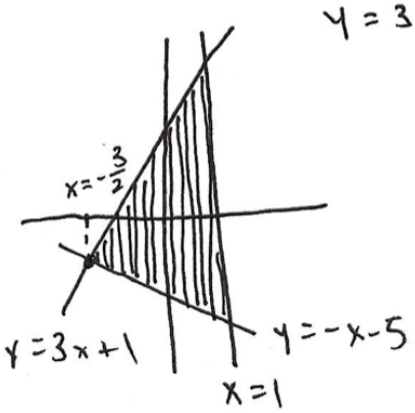
$\cos x = \frac{1}{2}$

$x = \frac{\pi}{3}, -\frac{\pi}{3}$ for $x \in [-\frac{\pi}{3}, \frac{\pi}{3}]$

$$A = \int_{-\pi/3}^{\pi/3} (8\cos x - \sec^2 x)\,dx$$
$$= (8\sin x - \tan x)\Big|_{-\pi/3}^{\pi/3}$$
$$= \left(8\sin\frac{\pi}{3} - \tan\frac{\pi}{3} - \left(8\sin\left(-\frac{\pi}{3}\right) - \tan\left(-\frac{\pi}{3}\right)\right)\right)$$
$$= 2\left(8\sin\frac{\pi}{3} - \tan\frac{\pi}{3}\right)$$
$$= 2\left(8 \cdot \frac{\sqrt{3}}{2} - \sqrt{3}\right)$$
$$= \boxed{6\sqrt{3}}$$

ex 3: Find the area of the region bounded by the lines:

$y = 3x + 1, \quad y = -x - 5, \quad x = 1$



Intersect when

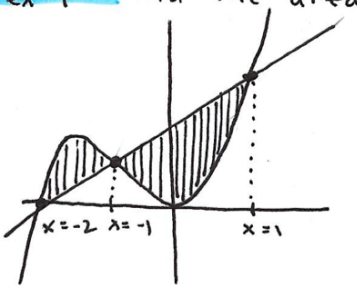
$3x + 1 = -x - 5$

$4x = -6$

$x = -\frac{3}{2}$

$$A = \int_{-3/2}^{1} [(3x+1) - (-x-5)]\,dx$$
$$= \int_{-3/2}^{1} (4x + 6)\,dx$$
$$= \left(\cancel{4}^{2} \cdot \frac{x^2}{\cancel{2}} + 6x\right)\Big|_{-3/2}^{1}$$
$$= (2+6) - \left(\frac{9}{2} - 9\right)$$
$$= \boxed{12\tfrac{1}{2}}$$

ex 4: Find the area of the region between $y = x^3 + 2x^2$ and $y = x + 2$.



Intersect when

$x^3 + 2x^2 = x + 2$

$x^3 + 2x^2 - (x+2) = 0$

$x^2(x+2) - (x+2) = 0$

$(x^2-1)(x+2) = 0$

$(x-1)(x+1)(x+2) = 0$

$x = -2, -1, 1$

$$A = \int_a^b |f(x) - g(x)|\,dx$$

$$= \int_{-2}^{-1} \left[(x^3+2x^2) - (x+2)\right] dx + \int_{-1}^{1} \left[(x+2) - (x^3+2x^2)\right] dx$$

$$= \int_{-2}^{-1} (x^3 + 2x^2 - x - 2)\,dx + \int_{-1}^{1} (-x^3 - 2x^2 + x + 2)\,dx$$

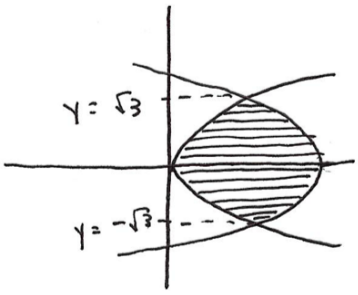
$$= \left(\frac{x^4}{4} + \frac{2x^3}{3} - \frac{x^2}{2} - 2x\right)\Big|_{-2}^{-1} + \left(-\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} + 2x\right)\Big|_{-1}^{1}$$

$$= \left(\frac{1}{4} - 4 - \frac{2}{3} + \frac{16}{3} - \frac{1}{2} + 2 + 2 - 4\right) + \left(-\frac{1}{4} + \frac{1}{4} - \frac{2}{3} - \frac{2}{3} + \frac{1}{2} - \frac{1}{2} + 2 + 2\right)$$

$$= \frac{5}{12} + 2\frac{2}{3}$$

$$= \boxed{3\frac{1}{12}}$$

ex 5: Find the area of the region between $x = y^2$ and $x = 6 - y^2$.



Intersect when

$y^2 = 6 - y^2$

$2y^2 = 6$

$y^2 = 3$

$y = \pm\sqrt{3}$

$$A = \int_{-\sqrt{3}}^{\sqrt{3}} (6 - y^2 - y^2)\,dy$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} (6 - 2y^2)\,dy$$

$$= \left(6y - \frac{2y^3}{3}\right)\Big|_{-\sqrt{3}}^{\sqrt{3}}$$

$$= 6(\sqrt{3} + \sqrt{3}) - \frac{2}{3}(3\sqrt{3} + 3\sqrt{3})$$

$$= \boxed{8\sqrt{3}}$$